

Finite-Dimensional p-Excess Invariance for Approximately Dual p-ASFs

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Source and Question

The source paper is K. Mahesh Krishna and P. Sam Johnson, *Approximately Dual p-Approximate Schauder Frames*, arXiv:2110.10121. Problem 3.3 asks whether, for two approximately dual p-approximate Schauder frames

$$(\{f_n\}_n, \{\tau_n\}_n), \quad (\{g_n\}_n, \{\omega_n\}_n)$$

for a Banach space $\mathcal{X}$, one always has

$$\text{p-Exc}(\{f_n\}_n, \{\tau_n\}_n) = \text{p-Exc}(\{g_n\}_n, \{\omega_n\}_n).$$

Here

$$\text{p-Exc}(\{f_n\}, \{\tau_n\}) = \sup \left\{ |M| : M \subseteq \mathbb{N}, \overline{\text{span}}\{\tau_n : n \notin M\} = \mathcal{X}, \overline{\text{span}}\{f_n : n \notin M\} = \mathcal{X}^* \right\}.$$

Proof Intuition

In finite dimension, a p-ASF is already controlled by arbitrarily large finite chunks. Since the analysis map has finite-dimensional domain, the coordinate truncations converge uniformly on its range in ℓ^p . Thus large partial frame operators are invertible. For a finite invertible partial frame operator, Cauchy-Binet forces at least one d -element set of indices whose vectors span $\mathcal{X}$ and whose functionals span $\mathcal{X}^*$. All other indices in that finite chunk may be deleted while those d indices are left behind. Letting the chunk grow gives infinite p-excess.

Finite Matrix Lemma

Lemma 1. *Let $\dim \mathcal{X} = d < \infty$, and let $(f_i, \tau_i)_{i=1}^N$ be a finite p-ASF analogue on $\mathcal{X}$, meaning that the operator*

$$Sx = \sum_{i=1}^N f_i(x) \tau_i$$

is invertible. Then

$$\text{p-Exc}((f_i)_{i=1}^N, (\tau_i)_{i=1}^N) = N - d.$$

Proof. No deletion set can have cardinality larger than $N - d$, since fewer than d remaining vectors cannot span $\mathcal{X}$, and fewer than d remaining functionals cannot span $\mathcal{X}^*$.

Choose a basis of $\mathcal{X}$ and the dual coordinate basis of $\mathcal{X}^*$. Put the vectors τ_i as the columns of a $d \times N$ matrix T , and put the functionals f_i as the rows of an $N \times d$ matrix F . The matrix of S is TF , hence $\det(TF) \neq 0$. By Cauchy-Binet,

$$\det(TF) = \sum_{\substack{I \subseteq \{1, \dots, N\} \\ |I|=d}} \det(T_I) \det(F_I),$$

where T_I is the $d \times d$ matrix of columns indexed by I , and F_I is the $d \times d$ matrix of rows indexed by I . Therefore some I has both determinants nonzero. The vectors $\{\tau_i : i \in I\}$ span $\mathcal{X}$, and the functionals $\{f_i : i \in I\}$ span $\mathcal{X}^*$. Deleting the complementary set $\{1, \dots, N\} \setminus I$, of cardinality $N - d$, is therefore allowed. This proves the reverse inequality. $\square$

Finite-Dimensional Countable Case

Theorem 1. *Let $1 \leq p < \infty$, let $\dim \mathcal{X} = d < \infty$, and let $(\{f_n\}_{n=1}^\infty, \{\tau_n\}_{n=1}^\infty)$ be a p -ASF for $\mathcal{X}$. Then*

$$\text{p-Exc}(\{f_n\}, \{\tau_n\}) = \infty.$$

Proof. Let $\theta_f : \mathcal{X} \rightarrow \ell^p$ and $\theta_\tau : \ell^p \rightarrow \mathcal{X}$ be the analysis and synthesis operators. The frame operator

$$S = \theta_\tau \theta_f$$

is invertible. Let $P_N : \ell^p \rightarrow \ell^p$ be the coordinate projection onto the first N coordinates and set

$$S_N = \theta_\tau P_N \theta_f = \sum_{n=1}^N f_n(\cdot) \tau_n.$$

Because $\mathcal{X}$ is finite dimensional, the image of its unit ball under θ_f is compact in ℓ^p . Since $P_N y \rightarrow y$ for each $y \in \ell^p$, the convergence is uniform on this compact image. Hence

$$\|S_N - S\| \leq \|\theta_\tau\| \|(P_N - I)\theta_f\| \rightarrow 0.$$

For all sufficiently large N , S_N is invertible. Applying Lemma 1 to the finite family $(f_n, \tau_n)_{n=1}^N$, we obtain a set $I_N \subseteq \{1, \dots, N\}$ with $|I_N| = d$ such that $\{\tau_n : n \in I_N\}$ spans $\mathcal{X}$ and $\{f_n : n \in I_N\}$ spans $\mathcal{X}^*$. Therefore the deletion set

$$M_N = \{1, \dots, N\} \setminus I_N$$

is admissible for the original countable p -ASF, because the remaining countable family still contains the spanning subfamilies indexed by I_N . Thus

$$\text{p-Exc}(\{f_n\}, \{\tau_n\}) \geq |M_N| = N - d$$

for arbitrarily large N , and the p -excess is infinite. $\square$

Corollary 1. *Problem 3.3 has an affirmative answer for finite-dimensional Banach spaces. Indeed, if $\dim \mathcal{X} < \infty$, any two p -ASFs for $\mathcal{X}$, approximately dual or not, have the same p -excess, namely ∞ . For the finite-index analogue with N atoms, any two finite p -ASFs for $\mathcal{X}$ have p -excess $N - \dim \mathcal{X}$.*

Limitations

The proof does not settle the infinite-dimensional case of Problem 3.3. In infinite dimension, finite coordinate truncations need not approximate the frame operator in operator norm, and the Cauchy-Binet common-subset argument has no direct infinite-dimensional replacement. The additional requirement that the remaining functionals span $\mathcal{X}^*$, not only that the remaining vectors span $\mathcal{X}$, is the point that still requires a genuinely infinite-dimensional argument or a counterexample.

Novelty Check

Local run indexes and attempts contained no existing result packet for arXiv:2110.10121 or Problem 3.3. Web searches used the phrases

- "p-Exc" "approximately dual",
- "Approximately Dual p-Approximate Schauder Frames" "excess",
- "p-excess" "p-approximate Schauder frames".

These searches found the source paper and the Hilbert-frame excess paper of Bakic and Beric, but no later paper resolving the p-ASF problem. The finite-dimensional result above is therefore recorded as a run-generated partial result, subject to human review.

References

- [1] K. Mahesh Krishna and P. Sam Johnson, *Approximately Dual p-Approximate Schauder Frames*, arXiv:2110.10121, 2021.
- [2] Damir Bakic and Tomislav Beric, *On excesses of frames*, Glasnik Matematički Serija III 50(70) (2015), no. 2, 415–427; arXiv:1403.0632.